

Descriptive Statistics - central tendency and dispersion _ Data Normalization

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Descriptive Statistics

- It focuses on summarizing and describing the main features of a dataset by providing information like:
- what are their average values, what are the most common values,
- how spread out they are, and their overall distribution.
- **It involves:**
- Measure of Central Tendency
- Measure of Dispersion



Central Tendency Measure

- *Central tendency is a statistical concept that refers to the tendency of data to cluster around a central value or a typical value.*
- It identifies a single value as representative of an entire data distribution. In other words, the [central tendency](#) is a way of describing the centre or midpoint of a dataset. The three most common measures of central tendency are:
 - [Mean](#)
 - [Median](#)
 - [Mode](#)



Mean – Median - Mode

Mean

The mean, also known as the average, is calculated by summing up all the values in a dataset and dividing the sum by the total number of values. It is sensitive to extreme values, making it susceptible to outliers.

Median

The median is the **middle value in a dataset when the values are arranged in ascending or descending order**. If there is an even number of values, the median is the average of the two middle values.

Mode

- The mode is the value that occurs most frequently in a dataset. It is particularly useful for categorical data but can also be applied to numerical data. A dataset may have one mode (unimodal), two modes (bimodal), or more than two modes (multimodal).



Dispersion

- **What is Dispersion?**
- Dispersion, also known as **variability or spread**, measures the extent to which individual data points deviate from the central value. It provides information about the spread or distribution of data points in a dataset.
- Common measures of dispersion include
 - [Range](#)
 - [Variance](#)
 - [Standard Deviation](#)
 - [Interquartile Range \(IQR\)](#)



Dispersion

Range

- In statistics, the range refers to the difference between the highest and lowest values in a dataset. It provides a simple measure of **variability, indicating the spread of data points**. The range is calculated by subtracting the lowest value from the highest value.
- For example, in a dataset {4, 6, 9, 3, 7}, the range is $9 - 3 = 6$.

Variance

- Variance is a statistical measure that quantifies the **amount of variation or dispersion of a set of numbers from their mean value**. Specifically, variance is defined as the expected value of the squared deviation from the mean.



Dispersion

- **Standard Deviation**

- Standard deviation is a measure of the amount of variation or dispersion of a set of values from the mean value. It is calculated as the square root of the variance, which is the average squared deviation from the mean.

- **Interquartile Range (IQR)**

- Interquartile Range (IQR) is a measure of statistical dispersion that represents the middle 50% of a data set. It is calculated as the difference between the 75th percentile (Q_3) and the 25th percentile (Q_1) of the data i.e., $IQR = Q_3 - Q_1$.



Data Normalization

- It is not necessary for all datasets in a model.
- It is required only **when features of machine learning models have different ranges**.
- This usually entails rescaling the features to a standard range, which is typically **between 0 and 1**. Alternatively, the features can be adjusted to have a mean of 0 and a standard deviation of 1.



Approaches

Min- Max
Scaling

Z-Score
Normalization
(Standardisation)



Need For Data Normalization

- It guarantees that every feature contributes equally, preventing larger-magnitude features from overshadowing others.
- It enables **faster convergence of algorithms for optimization**, especially those that depend on gradient descent.
- Normalization improves the **performance of distance-based algorithms** like k-Nearest Neighbours.
- Normalization improves overall performance by **addressing model sensitivity problems in algorithms** such as Support Vector Machines and Neural Networks.



Min- Max Normalization:

- This method of normalizing data involves transforming the original data linearly.
- The data's minimum and maximum values are obtained, and each value is then changed using the formula that follows.

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$$



Min- Max Normalization:

- The formula works by subtracting the minimum value from the original value to determine how far the value is from the minimum.
- Then, it divides this difference by the range of the variable (the difference between the maximum and minimum values).

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$$



Min- Max Normalization:

- When the feature X is at its minimum, the normalized value (X'/X') is 0. This is because the numerator becomes zero.
- Conversely, when X is at its maximum, X'/X' is 1, indicating full-scale normalization.
- For values between the minimum and maximum, X'/X' ranges between 0 and 1, preserving the relative position of X within the original range.



Outliers

- Outliers are data points that deviate significantly from the overall distribution of the data.
- They can arise due to variability in the data or measurement errors and can adversely affect the performance of machine learning models.

Causes of Outliers

Data Entry Errors: Mistakes in manual data entry or system glitches.

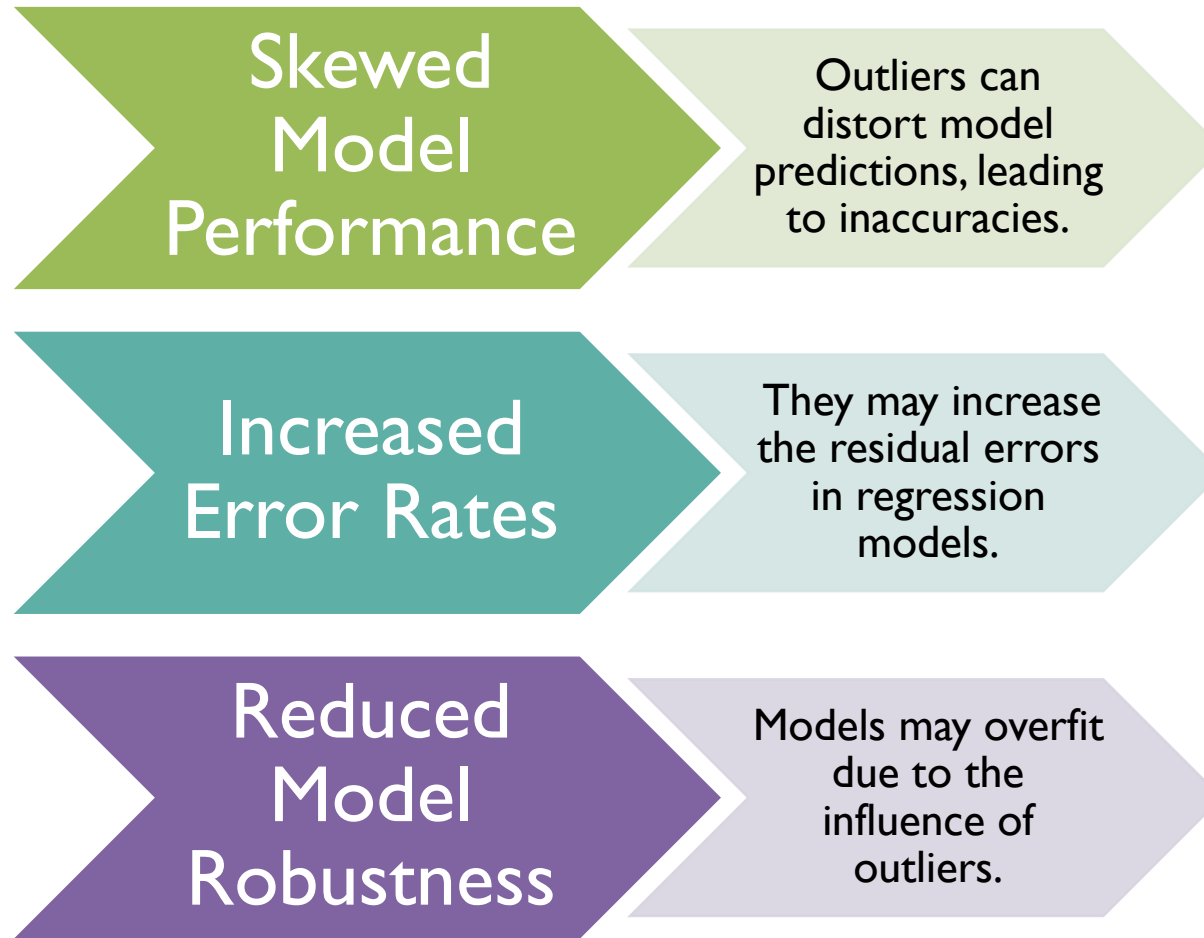
Measurement Errors: Errors in sensors or data collection tools.

Natural Variation: Genuine variations in the data distribution.

Experimental Conditions: Data points generated under different experimental settings.



Impact of Outliers



Types of Outliers

Univariate Outliers

- Outliers detected in one-dimensional data.

Multivariate Outliers

- Outliers found when considering relationships between multiple features.

Global Outliers

- Points that are far from the rest of the data.

Contextual Outliers:

- Points that are considered outliers only in a specific context.

Collective Outliers

- A group of data points that collectively deviate from the data distribution.

